

Assignment-1

L No13: Organism and Population

Section A

1. Which factor of an ecosystem includes plants, animals and microorganisms?

- a. **Biotic factor**
- b. Abiotic factor
- c. Direct factor
- d. Indirect factor

1. An assemblage of individuals of different species living in the same habitat and having functional interactions is

- a. **Biotic community**
- b. Ecological niche
- c. Population
- d. Ecosystem

2. Association between sea anemone and Hermit crab in gastropod shell is that of

- a. Mutualism
- b. **Commensalism**
- c. Parasitism
- d. Amensalism

3. Select the statement which explains best parasitism.

- a. One species is benefited.
- b. Both the species are benefited.
- c. One species is benefited, other is not affected.
- d. **One species is benefited, other is harmed.**

4. Growth of bacteria in a newly inoculated agar plate shows

- a. **exponential growth**
 - b. logistic growth
 - c. Verhulst-Pearl
 - d. zero growth
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5. **Ecology** is study of behaviour of animals in relation to their environment.

6. Organisms of same kind inhabiting a geographical area constitute **population**.

7. **Ethology** was studied for the relationship of animals and plants with their surroundings.

8. **Ecology** was study of relation between organisms to their environment.

9. **Environmental biology** gives emphasis on the functional or physiological interrelationships between the organism and their surroundings.

10. All the ecosystems on earth constitute **Biosphere**.

11. **Habitat** is a place or the set of environmental conditions around the organism to which it must adapt to survive and prosper.

12. **Biome** constitutes a large regional terrestrial unit delimited by a specific climatic zone having major vegetation zone (plant communities) and the associated fauna.

13. **Niche** is used to denote the functional role played by an organism in its environment.

14. In colloquial language habitat is a postal address while niche is the niche of an organism.
15. The immediate surrounding of an organism, sometimes also referred to as microhabitat.
16. Spatial niche deals with the physical space occupied by the organisms.
17. Trophic niche is on the basis of trophic level of an organism in a food chain.
18. Multidimensional or hypervolume niche considers number of environmental factors (both biotic and abiotic), the resulting space will be a hypervolume; not something that can be perceived by the human mind.
19. Fundamental Niche is the niche in the absence of all competitors; this is highly improbable in nature.
20. Realized Niche is more realistic approach, in the presence of competition for the resources available in the habitat.
21. Only few organisms can tolerate and thrive in a wide range of temperatures, they are called Eurythermal.
22. A vast majority of organisms are restricted to a narrow range of temperatures, they are called Stenothermal.
23. Some organisms are tolerant for a wide range of salinities, they are called euryhaline.
24. Some organisms are restricted to a narrow range of salinities, they are called stenohaline.
25. The organisms try to maintain the constancy of its internal environment which is called homeostasis despite variations in the external environmental conditions.
26. In aquatic animals, the osmotic concentration of the body fluids changes with that of the ambient water osmotic concentration. These animals and plants are simply conformers.
27. The ability of seed to reduce their metabolic activities and going into a state of 'dormancy'.
28. Many desert plants have a thick cuticle on their leaf surfaces and have their stomata in deep pits to minimize loss of water through transpiration.
29. In the polar seas, aquatic mammals like seals have a thick layer of fat called blubber below their skin acting as an insulator to reduce loss of body heat
30. In animals, the organism, if unable to migrate may go into hibernation during winter e.g. polar bear.
31. Some snails and fish go into aestivation to avoid summer heat.
32. Desert plants have special photosynthetic pathway called CAM pathway.
33. Mammals from colder climates generally have shorter snout, ears, tail and limbs to minimize the loss of body heat this is Allen's rule.
34. Natality is the birth rate of a population
35. Mortality is the death rate of a population
36. Crude birth rate is used when calculating population size (number of births per 1000 population/year).
37. Specific birth rate is used relative to a specific criterion such as age.
38. Realized natality is the number of births when environmental pressures come into play.
39. Absolute natality is the number of births under ideal conditions (with no competition, abundance of resources such as food and water, etc.).
40. The sex ratio of the population affects and is reciprocally affected by birth, death, immigration, and emigration rates

41. The males and females are in a ratio of 1:1 is generally the most common **evolutionary stable strategy**
42. If the age distribution (per cent individuals of a given age or age group) is plotted for the population, the resulting structure is called an **age pyramid**
43. **Immigration** is the number of individuals of the same species that have come into the habitat from elsewhere during the time period under consideration
44. **Emigration** is the number of individuals of the population who left the habitat during the time period.
45. Ideally, when resources in the habitat are unlimited, each species has the ability to fully realize its innate potential to grow in numbers. Then the population grows in an **exponential or geometric** proportion
46. In population interactions both the species are benefited in **mutualism** and both are harmed in **competition**.
47. In **parasitism** and **predation** only one species benefits
48. The interaction where one species is benefitted and the other is neither benefitted nor harmed is called **commensalism**
49. **Gause's Competitive Exclusion Principle** states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually
50. In interspecific competition with sufficient resources, species facing competition will evolve mechanisms that promote co-existence rather than exclusion. One such mechanism is **resource partitioning**.
51. **Brood parasitism** in birds is a fascinating example, in which the parasitic bird lays its eggs in the nest of its host bird and lets the host bird incubate them.
52. Some species of insects and frogs are cryptically coloured **comouflaged** to avoid being detected easily.

Notes:

1. Define the following terms :

a. Commensalism:

This is the interaction in which one species benefits and the other is neither harmed nor benefited.

b. Parasitism

This is an interaction in which a majority of the parasites harm the host. They may reduce the survival, growth and reproduction of the host and may lead to death of the host, thus reducing its population density.

c. Camouflage

Some species of insects and frogs are cryptically coloured (camouflaged) to avoid being detected easily.

Example: The Monarch butterfly is highly distasteful to its predator (bird) because of a special chemical present in its body. Interestingly, the butterfly acquires this chemical during its caterpillar stage by feeding on a poisonous weed.

Give one example for each :

a. Mutualism :-

Ans:- 1. Lichens represent an intimate, mutualistic relationship between a fungus and photosynthetic algae or cyanobacteria.

2. The most spectacular and evolutionarily fascinating examples of mutualism are found in plant-animal relationships. Plants need the help of animals for pollinating their flowers and dispersing their seeds. Animals obviously have to be rewarded in the form of pollen and nectar for pollinators and juicy and nutritious fruits for seed dispersers.

b. Interspecific competition

Ans:-

1. In interspecific competition with sufficient resources, species facing competition will evolve mechanisms that promote co-existence rather than exclusion.

2. One such mechanism is 'resource partitioning'. If two species compete for the same resource, they could avoid competition by choosing different times for feeding.

3. A species whose distribution is restricted to a small geographical area because of the presence of a competitively superior species, is found to expand its range when the competing species is removed.

c. Give an example of an animal which is introduced in an environment intentionally and there is no predator for the same?

Ans:-When certain exotic species are introduced accidentally or intentionally into a new geographical area, they become invasive and start spreading rapidly due to absence of natural predator, e. g. zebra mussels in the intertidal zone of North America.

4. Name the type of association:

- a. **Clown fish and sea anemone :-** Commensalism
- b. **Crow feeding the hatchling of Koel :-** Brood Parasitism
- c. **Humming birds and host flowering plants :** Mutualism
- d. **Cattle egrets always forage close to cattle,;-** Commensalism
- e. **An orchid growing as an epiphyte on a branch of mango tree.:-** Commensalism
- f. **Sparrow eating any seed:** - Predation
- g. **Grazers are animals eating plants:** - Predation
- h. **Lice on humans and ticks on dogs:-** Ectoparasite
- i. **Marine fish are infested with copepods:-** Ectoparasite
- j. ***Cuscuta*, that is commonly found growing on hedge plants:-** Parasitic Plant
- k. **In shallow creeks on the west coast of Mumbai, visiting flamingos and resident fish:-** Competition
- l. **Lichens:-** Mutualism
- m. **Flower and Insect:** Mutualism

5. What is the ecological process behind the biological control method of managing with pest insects?

Ans:-

- 1. The basis of various **biological control methods** is on the concept of predation.
- 2. Predation is a **biological** interaction between the predator and the prey, whereby the predator feeds on the prey.
- 3. Hence, the predators regulate the population of preys in a habitat, thereby helping in the **management of pest insects**
- 4. E.g. in absence of frogs, locusts increase in density and destroy large tracts of agricultural lands.
- 5. Biological control, methods adopted in agricultural pest control are based on the ability of the predator to regulate prey population. Predators also help in maintaining species diversity in a community, by reducing the intensity of competition among competing prey species.

Q. 3 Short answer questions.

1. How is the dormancy of seeds different from hibernation in animals?

Ans:-1. In plants,(during stress conditions) seeds serve as means to tide over periods of stress; they germinate to form new plants under favourable moisture and temperature conditions. They do so by reducing their metabolic activity and going into a state of '**dormancy**'.

2. In animals, the organism (in stress conditions), if unable to migrate may go into **hibernation** during winter e.g. polar bear.

3. Hibernation is also a state in which an animal slows down all metabolic activities like respiration, digestion, excretion etc. Process of reproduction is completely stopped.

4. In plants dormancy can occur both in dry season (sudden drop of water in environment) while Hibernation is observed in animals to escape from winter (extreme cold season) alone.

5. If a marine fish is placed in a fresh water aquarium, will it be able to survive? Give reason.

Ans:- No it can't survive

Reasons:-

1. The dissolved salt concentration (measured as salinity in parts per thousand), is less than 5ppt in fresh waters of streams , lakes and rivers, and 30-35ppt in the seas and oceans.
2. It may go up to 100ppt in some hypersaline lagoons.
3. Some organisms are tolerant for a wide range of salinities (euryhaline) but others are restricted to a narrow range (stenohaline).
4. Hence, many fresh water animals cannot live for long in sea water and marine fish cannot survive in fresh water because of the osmotic problems, they would face.

6. Name important defense mechanisms in plants against herbivores.

Ans:-

1. For plants, herbivores are the predators.
 2. Plants therefore have evolved variety of morphological and chemical defenses against herbivores.
 3. Thorns (*Acacia*, Cactus) are the most common morphological means of defense.
 4. Many plants produce and store chemicals that make the herbivore sick.
 5. When chemicals/ produce are eaten, they inhibit feeding or digestion of predator and disrupt reproduction or even kill it.
 6. *Calotropis* growing in abandoned fields, produces highly poisonous cardiac glycosides and that is why you never see any cattle or goats browsing on this plant.
 7. A wide variety of chemical substances that we extract from plants on a commercial scale, (nicotine, caffeine, quinine, strychnine, opium, etc.,) are secondary metabolites produced by them actually as defences against grazers and browsers.
- 7. An orchid plant is growing on the branch of mango tree. How do you describe this interaction between the orchid and the mango tree?**

Ans: It is Commensalism

Description:

1. This is the interaction in which one species benefits and the other is neither harmed nor benefited.
2. An orchid growing as an epiphyte on a branch of mango tree, will get benefit while the mango tree derives no benefit

8. Distinguish between the following:

Habitat	Niche
A habitat is an area, where a species lives and interact with the	A niche is a concept, of how an organism lives or survives in the

other factors and prosper.	environmental conditions.
Habitat consists of numerous niches.	Niches do not contain such components.
Effect of temperature, rainfall and other abiotic factors.	Flow of energy from one organism to other through ecosystem.
Habitat supports numerous species at a time.	Niche supports a single species at a time.
Habitat is a physical place.	Niche is an activity performed by organisms.
Habitat is not species specific.	Niche is species specific.

a. Hibernation and Aestivation

Ans:-

Hibernation	Aestivation
Also known as “winter sleep”.	Also known as “summer sleep”.
Longer duration.	Short duration.
Animals look for a warm place to sleep.	Animals look for a moist, cool and shady place to sleep.
It prevents any internal body damage due to low temperatures.	It prevents excessive water loss and internal body damage due to high temperatures.
Hibernation takes place in warm and cold-blooded animals like bats, mammals, birds, etc.	Aestivation takes place in cold-blooded animals like snails, earthworms, frogs, etc.

b. Ectotherms and Endotherms

Ans:-

Ectotherms:
1. These are cold blooded poikilothermal animals.

2. They are unable to regulate their body temperature which changes with change of temperature of environment.
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3. These animals show hibernation (winter sleep) and aestivation (summer sleep), e.g., fish, frog, lizards

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Endotherms:

1. These are warm blooded homoeo-thermal animals.

2. They maintain their body temperature in every condition.

3. These two activities rarely occur in these animals, e.g., Aves, mammals

Parasitism and Mutualism

MUTUALISM VERSUS PARASITISM

MUTUALISM	PARASITISM
Interactions between organisms of two different species, in which each organism benefits from the interaction in some way	A type of symbiotic relationship, or long-term relationship between two species, where one member, the parasite, gains benefits that come at the expense of the host member
Occurs between two non-parasitic species	Occurs between the host and the parasite
Two organisms need each other	Parasite needs the host, but the host doesn't need the parasite
Not very specific	A kind of specific relationship
Ex: relationship between bees and flowers, digestive bacteria and humans, oxpeckers and zebras, clownfish and sea anemone	Ex: relationship between mosquito and human, lice on human, tapeworms in cow, and Cuscuta in plants
	Visit www.PEDIAA.com

1. Write a short note on

a. Adaptations of desert animals/ Behavioural adaptations in animals

Ans:

1. Some organisms show behavioural responses to cope with variations in their environment.
2. Desert lizards manage to keep their body temperature fairly constant by behavioural adaptations.
3. They bask in the sun and absorb heat, when their body temperature drops below the comfort zone, but move into shade, when the ambient temperature starts increasing.
4. Some species burrow into the sand to hide and escape from the heat.

b. Adaptations of plants to water scarcity

Ans:

1. Many desert plants have a thick cuticle on their leaf surfaces and have their stomata in deep pits to minimize loss of water through transpiration.
2. They also have a special photosynthetic pathway (CAM - Crassulacean acid metabolism) that enables their stomata to remain closed during daytime.
3. Some desert plants like *Opuntia*, have their leaves reduced (modified) to spines and the photosynthetic function is taken over by the flattened stems.

1. Define the following:-

Ans:

Population:-Organisms of same kind inhabiting a geographical area constitute **population**

Community:- Several populations of different species in a particular area constitute **community** that interact with one another in several ways.

Biome : It constitutes a large regional terrestrial unit delimited by a specific climatic zone having major vegetation zone (plant communities) and the associated fauna

Ethology - The term was coined by Hilaire (1854) but was popularised by W. M. Wheeler (1902). The term denotes (speaks for) the study of behaviour of animals in relation to their environment.

Ecology - The term was introduced by E. Haeckel (1865) for the relationship of animals and plants with their surroundings.

Bionomics - Lankester (1890) coined this term for the study of relation between organisms to their environment.

Environmental biology (modern ecology) - The term was introduced in 20th century (G. L. Clarke 1964, Odum 1969) giving emphasis on the functional or physiological interrelationships between the organism and their surroundings.

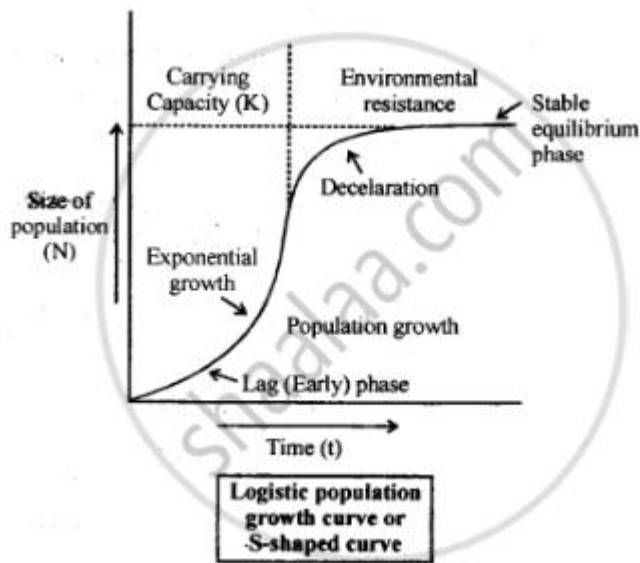
Biosphere - All the ecosystems on earth constitute biosphere.

Q. 4 Long answer questions.

1. With the help of suitable diagram describe the logistic population growth curve.

Ans:-

2. competition between individuals for limited resources will weed out the 'weaker' ones.
 3. Only the 'fittest' individuals will survive and reproduce.
 4. In nature, a given habitat has enough resources to support a maximum possible number, beyond which no further growth is possible.
 5. Let us call this limit as nature's carrying capacity (K) for that species in that habitat.
- A population growing in a habitat with limited resources show initially a lag phase, followed by phases of acceleration and deceleration and finally an asymptote, when the population density reaches the carrying capacity. A plot of population density (N) in relation to time (t) results in a sigmoid curve. This type of population growth is called Verhulst-Pearl Logistic Growth.
6. Since resources for growth for most animal populations, are finite and become limiting sooner or later, the logistic growth model is considered a more realistic one.



7. Enlist and explain the important characteristics of a population

Ans:-

1. Basic physical characteristics of population are: -

- a) **size and density**
- b) natality,
- c) mortality,
- d) immigration,
- e) emigration,
- f) age pyramids,
- g) expanding population,
- h) population growth forms and
- i) biotic potential.

a) **Size & Density:**

Size speaks for the number of individuals in a population while density tells us number of individuals present per unit space, in a given time.

b) **Natality:** is the birth rate of a population

Crude birth rate is used when calculating population size (number of births per 1000 population/year), whereas **specific birth rate** is used relative to a specific criterion such as age. E.g. If in a pond, there were 200 carp fish last year and through reproduction 800 new fish are added, taking the current population to 1000, we calculate the birth rate as $800/200 = 4$ offspring per carp per year.

Absolute Natality : the number of births under ideal conditions (with no competition, abundance of resources such as food and water, etc.).

Realized Natality : the number of births when environmental pressures come into play. It must be remembered, that absolute natality will be always more than realized natality.

c) **Mortality** is the death rate of a population. Mortality rate or death rate, is a measure of the number of deaths (in general, or due to a specific cause) in a particular population, in proportion to the size of that population, per unit of time.

Absolute Mortality : the number of deaths under ideal conditions (with no competition, abundance of resources such as food and water, etc.).

• **Realized Mortality** : the number of deaths when environmental pressures come into play.

d) **Immigration**(I) is the number of individuals of the same species that have come into the habitat from elsewhere during the time period under consideration.

e) **Emigration** (E) is the number of individuals of the population who left the habitat during the time period.

So, if N is the population density at time 't', then its density at time 't +1' can be calculated as,

$$N_{t+1} = N_t + [(B + I) - (D + E)]$$

It must be remembered that absolute mortality will be always less than realized mortality.

f) **Sex ratio:**

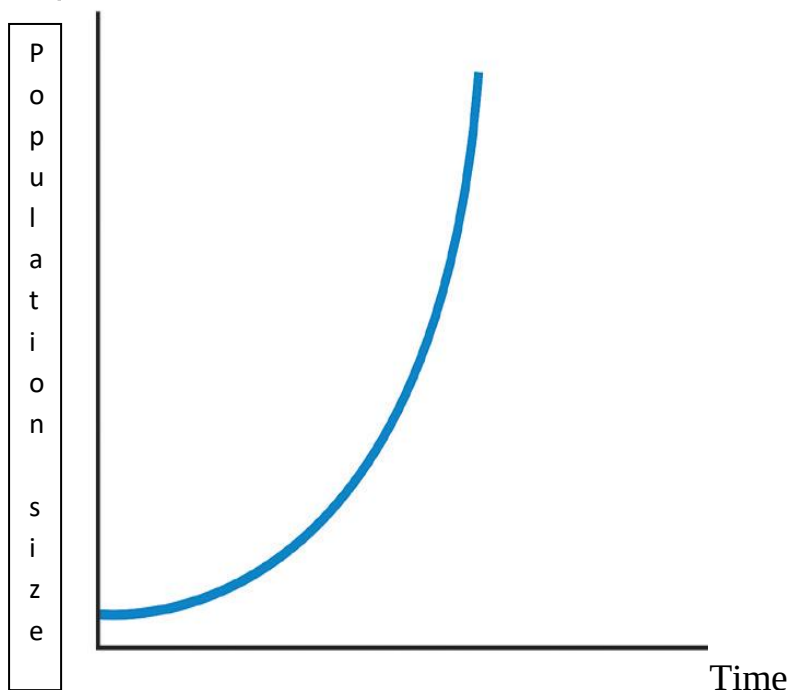
It is measured as the ratio of the number of individuals of one sex to that of the other sex. The males and females in a ratio of 1:1 is generally the most common **evolutionary stable strategy (ESS)**.

g) **Age distribution and Age pyramid:**

A population is composed of individuals of different ages. If the age distribution (per cent individuals of a given age or age group) is plotted for the population, the resulting structure is called an **age pyramid**. For the purpose of simplicity, the entire population is divided into three age groups as **Pre-Reproductive** (age 0-14 years), **Reproductive** (age 15-44 years) and **Post-reproductive** (45-85+ years).

h) **Expanding population.**

Resources like food, space are essential for any growth of a population. Ideally, when resources in the habitat are unlimited, each species has the ability to fully realize its innate potential to grow in numbers. Then the population grows in an exponential or geometric proportion.



i) **Population growth forms** are as follows:-

Rapid Growth e.g. Kenya, Nigeria

Slow Growth e.g. United States, Australia

Zero Growth e.g. Denmark, Italy

Negative Growth e.g. Germany, Hungary

APPLICATION BASED QUESTIONS

1. What is an ecosystem?

Ans:- An ecosystem is a community of living organisms in conjunction with the nonliving components of their environment, interacting as a system

2. What are the abiotic and biotic components of an ecosystem?

Ans:-Abiotic factors are non living things like water, air, soil, sunlight, and minerals.

Biotic factors are living things like animals, birds, plants, fungi, and other similar organisms

3. Give names of eurythermal and stenothermal animals and plants?

- **Ans:-**Eurythermal organisms are those which can tolerate a wide range of temperature. Examples: Cat, Dog, Man Tiger.
- Eurythermal plants: roses, daisies, some vegetables and fruits etc.
- **(stenothermal organism)**: a vast majority of them are restricted to a narrow range of temperatures
- Stenothermal plants: bouganvillea, orchids, frangipani, some fruits, Croton etc
- Stenothermal animals include a penguin, python, crocodile, etc

4. What will be the effect of increasing global temperatures on the different habitats and the organisms found in those habitats?

Ans:-

- i) **Habitat destruction** (also termed **habitat loss** and **habitat reduction**) is the process by which a natural habitat becomes incapable of supporting its native species.
- ii) The organisms that previously inhabited the site are displaced or die, thereby reducing biodiversity and species abundance

5. Give examples of an animal and plant that can survive in fresh water as well as sea water?

Ans:-Plant:- A good example of this is furnished by Polysiphonia violacea. ... nately for some hours each day to salt water and to fresh water

- **Animal:** Euryhaline organisms are able to adapt to a wide range of salinities.
- *An example of a euryhaline fish is the molly (Poecilia sphenops) common molly fish which can live in fresh water, brackish water, or salt water.*
- The green crab (*Carcinus maenas*) is an example of a euryhaline invertebrate that can live in salt and brackish water

6. What is the source of energy for the life in deep ocean trenches where sunlight does not reach?

- **Ans:-**Even without sunlight, most life in the deep sea is still dependent on the sun.
- The dead organic matter on which these animals and bacteria feed ultimately comes from phytoplankton and other plants and organisms that harness sunlight to create energy in a process called photosynthesis

7. What is homeostasis?

Ans:- In biology, homeostasis is the state of steady internal, physical, and chemical conditions maintained by living systems.

8. Why do animals need to maintain homeostasis?

- i. **Ans:** To survive and flourish in any environment, organisms must adapt to the changes in the environment. **Homeostatic** mechanisms are essential for **animals**, as they regulate and **maintain** the organism in optimum conditions, even when they face adverse environmental conditions.

9. What are the adaptations in animals living under crushing pressure at great depths of ocean?

Ans:-

- i) The animals have a developed ways to cope with the deep ocean's intense water pressure.
- ii) This force, which is the result of the weight of a column of water pushing down, is so strong in the bathypelagic zone that it would crush anything with air in it.
- iii) That would be a problem for most fish, since many of them have a gas-filled organ called a swim bladder a large, thin-walled sac in some fishes that may function in several ways, e.g., as a buoyant float, a sound producer and receptor, and a respiratory organ. that helps them stabilize their bodies at different water depths
- iv) . But many deep-sea fish have adapted to life without a swim bladder.
- v) Without the bladders, deep-sea fishes don't have gas spaces in them that would cause them to get crushed.

11. What can be the reason behind the different reproductive strategies adopted by monocot plants like cereals/ pulses and dicot plants like mango?

12. Should an ideal parasite be able to thrive within the host without harming it?

13. Why didn't natural selection lead to the evolution of such totally harmless parasites?

14. Name the invasive species of plants and animals in India.

Remember

The instrument used to measure the hight of forest trees is called hypsometer.

World Environment day - 5thJune

World Population day - 11thJuly

World Earth day - 22ndApril

World Ozone day - 16thSeptember

Population Interaction

Remember

Positive Interaction	Negative Interaction
Mutualism	Predation
Syntrophism	Parasitism
Commensalism	Competition
Proto-cooperation	Ammensalism
Microbial Interaction	

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